

Technology Stack

Date	
Team ID	
Project Name	A Comprehensive of Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2 Templates	Provides a simple, responsive, and user-friendly interface for entering HDI parameters, selecting countries, and displaying prediction results dynamically.

2	Backend / Server-Side	Python 3.x, Flask Framework	Flask handles routing, request processing, user input validation, model prediction, and result rendering efficiently with lightweight architecture.
3	Database / Data Storage	CSV Dataset (hdi_dataset_clean.csv), Pickle (.pkl) Model File	The CSV file stores the cleaned HDI dataset, while the Pickle file stores the trained Linear Regression model for fast prediction without retraining
4	Cloud / Hosting / Deployment	Local Flask Development Server (localhost)	The application is executed locally for development and testing without requiring cloud deployment, making it simple and cost-effective.
5	Version Control & CI/CD	Git, GitHub	Git is used to track project changes, while GitHub stores the source code and supports collaboration and version management.

6	Third-Party APIs / Other Tools	Pandas, Scikit-learn, NumPy, Pickle, VS Code	Pandas processes the dataset, Scikit-learn trains and predicts using Linear Regression, NumPy performs numerical operations, Pickle serializes the trained model, and VS Code is used for development.